

Lab 9 - Sinusoids & Phasors

Objective

The objectives of this laboratory are to gain practical understanding and experience of AC circuits by:

- studying and measuring voltages in series RC, RL and RLC circuits
- expressing AC voltages using phasor notation
- calculating and measuring impedance
- observing impedance change as a function of the frequency of the applied source
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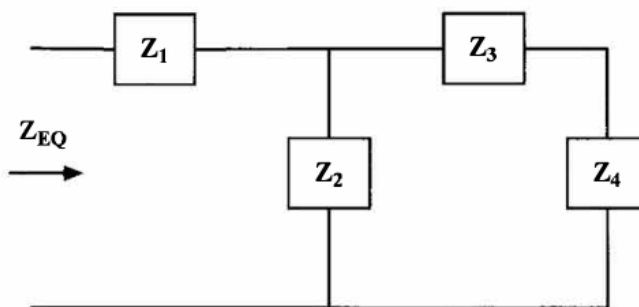
Introduction

Phasors

Phasors have magnitude and angle. One might think that a phasor is a vector but this is not the case. For one thing, voltage is not a vector but rather a scalar. We cannot apply curl, divergence, and dot product to voltages and get other voltages. Instead, a phasor is a way of expressing complex number that represent voltages and currents.

Impedance

The impedance of a circuit is a by-product of the method of undetermined coefficients that is used in solving for the steady state solution of a differential equation. Fortunately, all the rules we learned about resistance in DC circuits carry over to impedance in AC circuits. Impedance, in general, is a complex number composed of a real part and a j (imaginary) part. We write impedance as: $Z = R + jX$ with Z having units of ohms. X (called reactance) can be either a positive or negative number. It also has units of ohms. Consider the circuit below that is expressed with Z 's.



The Z_{EQ} is given by:

$$Z_{EQ} = Z_1 + [Z_2 || (Z_3 + Z_4)] = Z_1 + \frac{Z_2(Z_3 + Z_4)}{Z_2 + Z_3 + Z_4}$$

We see that the impedance is calculated just as we calculated resistance. There are two major differences between resistance and impedance. First, impedance has both a real part and a “ j ” part. Resistance has only a real part. Second, resistance does not change as the frequency of the applied voltage (or applied current) changes. The value of impedance does change as a function of the frequency of the applied voltage or current sources. This makes a world of difference in how circuits perform.

Procedure

Parts and Equipment needed:

- 100 Ω , 1k Ω , and 4.7k Ω resistors
- 0.1 μ F and 0.33 μ F capacitors
- 33mH and 10mH inductors
- XK700 Trainer
- Oscilloscope

Build and Measurements

- Build a series RC circuit using the 4.7k Ω resistor and 0.1 μ F capacitor connected to the sine wave source. Set the peak-to-peak source to 10V and vary the frequency according to the frequency table below. Measure the voltage and phase difference of V_C for each frequency.
- Similarly, build a series RL circuit using the 100 Ω resistor and 33mH inductor connected to the sine wave source. Set the peak-to-peak source to 10V and vary the frequency according to the table below. Measure the voltage and phase difference of V_L for each frequency.
- Finally, build a series RLC circuit using the 1k Ω resistor, 0.33 μ F capacitor, and 10mH inductor connected to the sine wave source. Set the peak-to-peak source to 10V and vary the frequency according to the table below. Measure the voltage and phase difference of both V_C and V_L for each frequency.

	RC Circuit		RL Circuit		RLC Circuit			
f (Hz)	V_C	ϕ_C	V_L	ϕ_L	V_C	ϕ_C	V_L	ϕ_L
100								
500								
1k								
5k								
10k								

Lab Report

- Title page
- Procedure: Write in your own words what you did in lab
- Data and Results: For each circuit:
 - Theoretical: Calculate the Z_T and V for each element expressed as phasors
 - Measured: Provide a typed/printed copy of your table of results
- Discussion: Did your measured values match your theoretical calculations? If not, what could be the source of the difference? What did you notice about the measured voltages as you varied the frequency? Did the output waveforms vary? If so, how?